

Danzel Yap

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Project Engineer & lifelong builder — from competition robotics to AI tools, driven by a passion for building and for artificial intelligence.

SUMMARY

Full-time Project Engineer managing structural foundation pours on the LA Convention Center expansion, and a builder at heart — from competition robotics to AI tools that give my team back a day each week. Equal parts field engineer and software developer, with a deep passion for AI and robotics.

EXPERIENCE

Webcor Craft, Project Engineer — Los Angeles, CA

2025 – PRESENT

- Manage **structural foundation pours** on the **LA Convention Center (LACC)** expansion as a full-time PE — coordinating concrete operations, embeds, takeoffs, and pour logistics from layout through placement.
- Proficient across core project-engineering workflows: **RFIs, submittals**, takeoffs, budgets, and look-ahead scheduling.
- Built an **AI in Construction Suite**, a purpose-built set of AI tools for the PM team that **saves every project engineer roughly a day each week** through tool tracking, reporting, and centralized workflows.
- Led the evaluation of an external AI vendor and recommended against adoption — **a call that saved the company \$75,000**.

Clune Construction Company, Project Management Intern — San Francisco, CA

JUNE 2024 – AUGUST 2024

- Analyzed drawings and developed project budgets and schedules in a team environment.
- Conducted AI research on practical construction applications — the seed of the AI in Construction Suite.

Cal Poly Pomona Formula E, Nuts & Bolts Lead (School project) — Pomona, CA

SEPTEMBER 2022 – JUNE 2024

- Led design and manufacture of mechanical systems — chassis, brakes, and driver ergonomics — for a competition electric vehicle.

K-12 STEM Center USA, Discover STEM Instructor (Part-time) — Claremont, CA

OCTOBER 2022 – MARCH 2026

- Taught robotics, Arduino, coding, and soldering to K-12 students; built and deployed an introductory AI course.

PROJECTS

AI in Construction Suite

2024 – PRESENT

- Built and deployed to a live project team, with purpose-built tools for live **pour tracking**, **embed installation tracking**, **tool inventory**, daily **attendance sign-in**, **look-ahead scheduling**, and concrete **cylinder break tracking**.
- Centralizes PM workflows and includes an AI chat assistant, integrated with the team's Google Workspace stack across web, Discord, and Teams.

FIRST Robotics Competition — California

NOVEMBER 2018 – JUNE 2021

- Designed and built competition robots with a multidisciplinary team; secured team funding and drove builds to deadline.

SKILLS

Concrete takeoff, RFIs, submittals, project management.

SOFTWARE

Python, JavaScript, Java / JavaFX, C#, MATLAB, VBA; LLM integration (Anthropic, OpenAI, Gemini), Firebase, Google Apps Script, React, Three.js; SolidWorks, Autodesk Fusion 360, Bluebeam Revu, Microsoft Office, Google Workspace.

EDUCATION

California State Polytechnic University, Pomona

B.S. Mechanical Engineering

Graduated Fall 2025

LANGUAGES

Fluent in Mandarin & English; conversational Cantonese.

INTERESTS

Robotics, applied AI, building (software & hardware), photography & videography (drone, gimbal, DSLR).